Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Mark Courage	Email	mark.courage@tno.nl
Institute Name	TNO (Netherlands Organisation for Applied Scientific Research)			
Address	Stieltjesweg 1, 2628 CK Delft, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Battery Safety Learning Using Fiber-Optic Sensing (FOS)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Measurement Instrument	Optical Measuring Instrument	Batteries	Manufacturing and Measurement/Evaluation Equipment
Concept of Proposed Technology	Battery safety regulations assume individual cell failures can occur and require battery packs to withstand such failures without hazardous propagation. While compliant designs meet regulatory standards, they often rely on conservative safety margins driven by limited insight into internal failure physics. This project proposes a 2-year, multi-phase program using electrically passive fiber-optic sensing (FOS) to characterize thermal and mechanical behavior during normal operation, abuse testing, and propagation events. The goal is not to prevent failure, but to understand failure physics, enabling optimized, regulation-compliant designs with reduced uncertainty, risk, cost, and iteration time.			

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Rationale for Joint R&D Project

Lithium-ion batteries are increasingly deployed in electric vehicles (EVs), energy storage systems (ESS), and industrial and mobility applications. However, thermal runaway, internal short circuits, and degradation-induced failure remain major safety risks. Several high-profile incidents have revealed that conventional point sensors (thermocouples, voltage taps) overlook or detect failures too late. Root-cause mechanisms are often poorly understood due to lack of internal, spatially resolved data. Safety improvements are reactive rather than predictive. This creates an urgent need for in-situ, high-resolution diagnostic tools that enable learning from early-stage failure related phenomena rather than post-incident analysis.

Battery fires and safety issues do not stop at national borders. Korean batteries which is the 3rd largest provider in the world, are deployed globally. EU regulators increasingly demand **explainability and traceability of safety performance.** OEMs need **early-warning systems** to meet insurance and liability requirements. A joint project aligns **Korean manufacturing excellence** with **European safety science.** The collaboration supports harmonisation of **future safety standards**. Korea–EU collaboration addresses global safety gaps and positions both parties ahead of regulatory tightening.

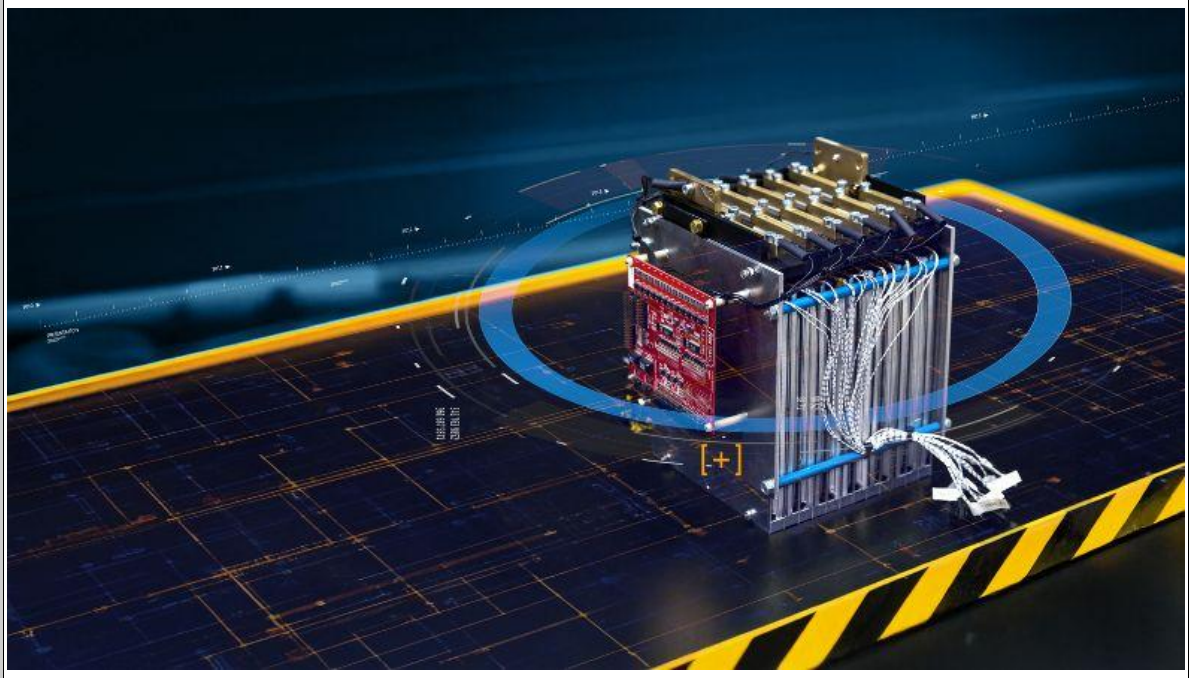

Figure 1 TNO accelerates electrification of logistics and construction through a deeper understanding of batteries.

The core innovation is not just installing fiber-optic sensors—but enabling:

- Learning from **thermal gradients, strain evolution, and anomaly patterns**
- Identification of **precursors to thermal runaway**
- Development of **digital safety fingerprints** for batteries

This requires:

- Controlled experiments
- Information about the internal process
- Cross-disciplinary interpretation
- Iterative feedback between design, sensing, and failure analysis

Such learning **cannot be achieved by isolated actors**.

Locked into fixed architectures, large battery corporations are often risk-averse in safety experimentation and focused on short-term commercialization. In contrast, South Korean SMEs can

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	prototype new sensing integration concepts rapidly and benefit most from **early failure detection and differentiation through safety**. Collaboration with SMEs thus enables **agile, learning-driven R&D**, which is critical for exploring FOS-based safety concepts. A joint R&D project between **TNO and South Korean SMEs** is **structurally necessary** to: • Move battery safety from reactive testing to proactive learning/design/development • Translate fiber-optic sensing from lab potential to industrial reality • Combine deep sensing science with real manufacturing practice • Address global safety, regulatory, and trust challenges in battery systems **Only through collaboration can fiber-optic sensing become a practical, scalable, and trusted tool for battery safety.**
Key Words	1. Battery safety intelligence 2. Fiber optic sensing 3. Early failure detection 4. Thermal runaway prevention 5. Photonics technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5-6)

R&D Objectives

Conventional battery monitoring relies mainly on point-based measurements of overall voltage, current, and surface temperature, which are insufficient to detect localized internal phenomena such as hotspots, internal short initiation, and mechanical deformation that precede thermal runaway. Recent studies show that thermal runaway is a progressive, spatially heterogeneous process, and early precursors often remain undetected by traditional sensors, leading to delayed warnings and limited learning from failures [1, 2].

Optical fiber sensors provide a promising solution, enabling distributed, *operando* measurement of internal temperature and strain with high spatial resolution, electrical insulation, and immunity to electromagnetic interference [3]. Fiber Bragg grating and distributed optical fiber sensing techniques have demonstrated earlier detection of abnormal thermal and mechanical behaviour compared to conventional approaches [4, 5]. Despite remaining challenges in integration and cost, optical fiber sensing offers a clear pathway toward predictive, learning-based battery safety monitoring.

During battery abuse testing and thermal-runaway propagation events, local temperatures can rapidly exceed 600–800 °C, accompanied by extreme thermal gradients, mechanical shock, flame exposure, and chemically aggressive environments. Conventional fiber-optic sensors used in battery monitoring (e.g., standard FBG or polymer-coated fibers) typically fail above 200–300 °C, primarily due to coating degradation, grating erasure, and mechanical breakage. This creates a critical technology gap: the inability to measure temperature and propagation dynamics during the most hazardous safety-critical phase of battery failure.

To enable sensing during thermal runaway and propagation, TNO holds sensor development Know-hows and engineering experience [6] with following fiber technologies:

- **Regenerated Fiber Bragg Gratings (R-FBGs)** capable of stable operation above **800 °C**
- **Femtosecond-laser-inscribed gratings** in pure silica fibers, avoiding hydrogen loading and thermal decay
- **Sapphire or single-crystal optical fibers** for ultra-high-temperature environments (>1000 °C)

These technologies allow temperature measurement beyond the limits of conventional telecom-grade fibers and are essential for capturing *peak temperatures and flame-front propagation* during abuse tests.

Furthermore, fiber survivability during battery failure requires development of:

- **Metallic coatings** (e.g., gold, aluminum) to replace polymer buffers to protect the optical fiber
- **Ceramic or mineral protective layers** resistant to flame, electrolyte vapor, and molten materials
- **Mechanically reinforced fiber routing** to withstand shock, rupture, and debris during cell venting

Such protection is critical to maintain sensor integrity during violent propagation events and post-failure measurement phases.

Thermal runaway propagation occurs on **milliseconds-to-seconds timescales**. Therefore, we will develop the sensor technology to support:

- **High temporal resolution (kHz-class sampling)** to resolve rapid temperature rise
- **Spatially distributed sensing** to track flame-front and heat-wave propagation across cell and modules

- Robust interrogation and multiplexing systems operating during high optical loss or partial fiber damage

Distributed optical fiber sensing techniques based on **Rayleigh or Raman scattering** are especially relevant for capturing spatial propagation dynamics in abuse tests. TNO's most advance instruments and equipment (example see Figure 2) will be applied and used in the research and development. Those distributed fiber sensing technology has in general a longer measurement time. Therefore, they will accompany the special high temperature FBGs for benchmarking and cross-calibration to increase the data accuracy and reliability. The benchmark will also yield insight for future product design when performance-cost trade-off evaluation is important.

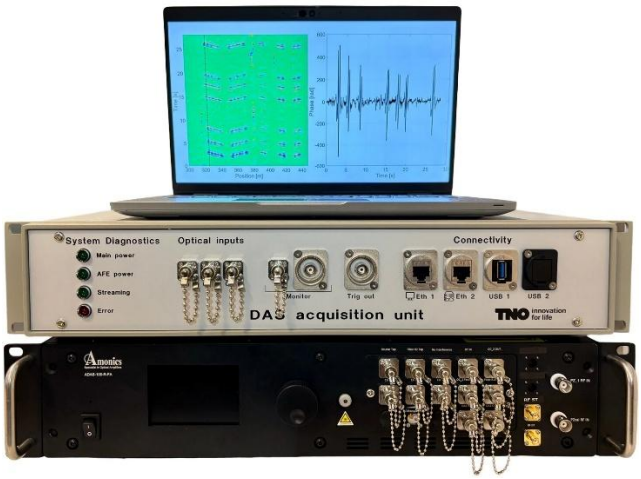

Figure 2 TNO distributed acoustic sensing acquisition unit.

The proposed technology developments enable:

- Direct measurement of **local peak temperatures and thermal gradients during runaway**
- Real-time mapping of **propagation pathways across cells and modules**
- Improved understanding of **fire spread mechanics and delay effectiveness**
- Data-driven validation of **propagation mitigation strategies** in battery design

These capabilities directly address gaps identified in current battery safety standards, where internal temperature evolution during propagation remains largely unknown.

Sensor integration and test-environment compatibility is another crucial technical development. TNO will build a battery pack using the battery cell/module provided by SMEs.

- Sensor placement strategies within modules and between cells
- Interfaces compatible with standardized abuse tests (over-temperature, nail penetration, propagation tests)
- Optical feedthroughs that preserve chamber integrity and fire safety

These developments enable **direct observation of failure evolution rather than post-test inference**.

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Figure 3 TNO test lab for next generation battery technology.

Perform testing under nominal and off-nominal conditions at TNO facility (Figure 3).

- Sensing data acquisition, processing
- incorporating AI-based safety monitoring using FOS spatial data.
- Controlled damage and needle penetrating test with high temperature FOS solutions

Using the data generated through fiber-optic sensing, the project will develop **diagnostic algorithms** that can:

- Identify abnormal sensing patterns linked to failure initiation
- Detect early indicators of internal short circuits, overheating, and/or deformation
- Distinguish normal aging behaviour from safety-critical anomalies

The overall R&D objectives are summarised as the followings:

- Develop **multipoint multiparameter fiber-optic sensing** (temperature/strain) **system** applicable to battery cells/modules.
- Develop **sensor integration techniques** compatible with SME-scale manufacturing constraints.
- Establish **high-resolution safety test data acquisition** under controlled abuse/failure scenarios.
- Develop **early-warning indicators and diagnostic algorithms** based on (quasi-)distributed sensing signals.
- Deliver **sensor-integrated prototypes** and **SME-focused integration guidelines** supporting commercialization and follow-up scale-up.

R&D Target Specifications

Performance Targets and KPIs are listed below:

Technical KPIs

- Distributed temperature measurement capability: spatially resolved profiling (minimum spatial resolution for critical locations: **2 mm**) over cell/module length
- Distributed strain/deformation monitoring: strain resolution **< 1 με**

	• Early-warning lead time: detection before critical safety event (target: 10 minutes before first venting.) • Integration robustness: sensor survivability during cycling and abuse tests • Data quality: synchronized dataset with full test metadata and repeatability evidence **Output KPIs** • Number of sensor-integrated prototypes: ≥ 10 cells and ≥ 3 modules • Number of controlled safety tests executed: ≥ 20 test cases • Diagnostic indicators developed: ≥ 3 validated indicators • Guideline document: integration + design rules (1 complete version)
R&D Plan	**WP0 – Project Management, Governance & Coordination** **Duration:** M1–M24 **Lead:** TNO **Participants:** All partners **Key Activities** • Project kick-off and mid-term review (M12) • Quarterly technical coordination meetings • IPR and data management enforcement • Safety governance for battery abuse testing **Roles** • **TNO:** Project coordinator, reporting lead, quality control • **Korean SMEs:** Governance participation, compliance, technical steering input **Key Deliverables** • D0.1 Project Management & IPR Plan (M2) • D0.2 Mid-term progress report (M12) • D0.3 Final project report (M24) **WP1 – Use-Case Definition & Safety Learning Objectives** **Duration:** M1–M3 **Lead:** South Korean SME(s) **Participants:** TNO **Key Activities** • Selection of battery type (cell or small module) • Definition of priority failure mechanisms • Mapping of learning questions (what must be detected *before* failure?) **Roles** • **Korean SMEs:** Provide battery designs, constraints, and application context • **TNO:** Translate needs into sensing and testing requirements **Key Deliverables** • D1.1 Use-case and safety learning requirements document (M3) **WP2 – Fiber-Optic Sensing Integration & Prototype Development** **Duration:** M3–M10 **Lead:** TNO **Participants:** Korean SMEs **Key Activities** • Selection of FOS modality (DFOS, FBG, DTS) • Sensor placement and embedding/pre-mounting strategy • Calibration and baseline characterization **Roles** • **TNO:** Sensor architecture, integration methodology, calibration

- **Korean SMEs:** Battery prototype fabrication, integration feasibility validation

Key Deliverables

- D2.1 Sensor integration guideline and design (M6)
- D2.2 Sensor-integrated battery prototypes (M10)

WP3 – Safety Testing & High-Resolution Data Generation

Duration: M8–M16

Lead: TNO

Participants: Korean SMEs

Key Activities

- Thermal, electrical, and mechanical abuse tests
- Real-time distributed temperature/strain acquisition
- Benchmark with conventional sensors

Roles

- **TNO:** Test execution, safety supervision, data acquisition
- **Korean SMEs:** Sample provision, test relevance validation, failure interpretation

Key Deliverables

- D3.1 Safety test dataset and experimental logbook (M16)
- D3.2 Pre-failure detection performance assessment (M16)

WP4 – Battery Safety Learning & Early-Warning Indicator Development

Duration: M14–M22

Lead: TNO

Participants: Korean SMEs

Key Activities

- Pattern recognition and anomaly detection
- Correlation of sensing signatures with failure mechanisms
- Development of safety indicators and alert logic

Roles

- **TNO:** Data analytics, learning framework, interpretation models
- **Korean SMEs:** Validation against real-world design and operational behaviour

Key Deliverables

- D4.1 Early-warning indicators for battery safety (M20)
- D4.2 Battery safety learning framework (M22)

WP5 – Industrial Validation, Exploitation & Roadmap

Duration: M18–M24

Lead: Korean SME(s)

Participants: TNO

Key Activities

- Evaluation of manufacturability and cost impact
- TRL assessment and gap analysis
- Exploitation and follow-up project definition

Roles

- **Korean SMEs:** Market relevance, productization strategy, customer alignment
- **TNO:** Support standardization pathways and future certification concepts

Key Deliverables

- D5.1 Industrial feasibility and exploitation roadmap (M24)
- D5.2 Recommendations for next-phase (TRL 6–7) development (M24)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (Netherlands Organisation for Applied Scientific Research)** contributes: • Deep expertise in **applied physics, sensing, data interpretation, and safety science** • Experience translating fundamental sensing technologies into **certifiable and deployable solutions** • Neutral, trusted position between industry, regulators, and researchers • Advanced test infrastructure and safety modelling capabilities
Expected R&R of Korean SME Partner	**South Korean battery SMEs contribute:** • Direct access to **real battery cells, modules, and packs** • Practical know-how in **manufacturing constraints, materials, and packaging** • Fast iteration cycles and strong willingness to experiment • Alignment with one of the world's most advanced battery ecosystems

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This international joint R&D will deliver: • sensor integrated battery cell/module prototypes using optical fiber sensing, including high temperature sensing concepts suitable for abuse/propagation tests, and • validated early warning indicators derived from distributed temperature/strain signatures to support "battery safety learning." Key outputs include integration guidelines for SME manufacturing, synchronized safety test datasets, and a prototype diagnostic logic (target TRL 5–6). Commercialization potential is high for Korean SMEs through new products/services such as safety enhanced modules, monitoring kits (sensor + analytics), and testing/diagnostic services for OEMs and ESS operators. Industrial impact includes improved understanding of thermal runaway initiation/propagation, reduced safety risk, and lower costs associated with recalls, downtime, and liability. Broader economic impact includes strengthened SME competitiveness, export readiness for stricter global safety expectations, and contribution to safer EV/ESS deployment supporting energy transition and sustainable growth.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization